
C Counting and Probability

This appendix reviews elementary combinatorics and probability theory. If you have a good background in these areas, you may want to skim the beginning of this appendix lightly and concentrate on the later sections. Most of this book's chapters do not require probability, but for some chapters it is essential.

Section C.1 reviews elementary results in counting theory, including standard formulas for counting permutations and combinations. The axioms of probability and basic facts concerning probability distributions form Section C.2. Random variables are introduced in Section C.3, along with the properties of expectation and variance. Section C.4 investigates the geometric and binomial distributions that arise from studying Bernoulli trials. The study of the binomial distribution continues in Section C.5, an advanced discussion of the “tails” of the distribution.

C.1 Counting

Counting theory tries to answer the question “How many?” without actually enumerating all the choices. For example, you might ask, “How many different n -bit numbers are there?” or “How many orderings of n distinct elements are there?” This section reviews the elements of counting theory. Since some of the material assumes a basic understanding of sets, you might wish to start by reviewing the material in Section B.1.

Rules of sum and product

We can sometimes express a set of items that we wish to count as a union of disjoint sets or as a Cartesian product of sets.

The **rule of sum** says that the number of ways to choose one element from one of two *disjoint* sets is the sum of the cardinalities of the sets. That is, if A and B are two finite sets with no members in common, then $|A \cup B| = |A| + |B|$, which follows from equation (B.3) on page 1156. For example, if each position on a car's license plate is a letter or a digit, then the number of possibilities for each position is $26 + 10 = 36$, since there are 26 choices if it is a letter and 10 choices if it is a digit.

The **rule of product** says that the number of ways to choose an ordered pair is the number of ways to choose the first element times the number of ways to choose the second element. That is, if A and B are two finite sets, then $|A \times B| = |A| \cdot |B|$, which is simply equation (B.4) on page 1157. For example, if an ice-cream parlor offers 28 flavors of ice cream and four toppings, the number of possible sundaes with one scoop of ice cream and one topping is $28 \cdot 4 = 112$.

Strings

A **string** over a finite set S is a sequence of elements of S . For example, there are eight binary strings of length 3:

000, 001, 010, 011, 100, 101, 110, 111.

(Here we use the shorthand of omitting the angle brackets when denoting a sequence.) We sometimes call a string of length k a **k -string**. A **substring** s' of a string s is an ordered sequence of consecutive elements of s . A **k -substring** of a string is a substring of length k . For example, 010 is a 3-substring of 01101001 (the 3-substring that begins in position 4), but 111 is not a substring of 01101001.

We can view a k -string over a set S as an element of the Cartesian product S^k of k -tuples, which means that there are $|S|^k$ strings of length k . For example, the number of binary k -strings is 2^k . Intuitively, to construct a k -string over an n -set, there are n ways to pick the first element; for each of these choices, there are n ways to pick the second

element; and so forth k times. This construction leads to the k -fold product $\underbrace{n \cdot n \cdots n}_{n \text{ times}} = n^k$ as the number of k -strings.

Permutations

A **permutation** of a finite set S is an ordered sequence of all the elements of S , with each element appearing exactly once. For example, if $S = \{a, b, c\}$, then S has 6 permutations:

$abc, acb, bac, bca, cab, cba$.

(Again, we use the shorthand of omitting the angle brackets when denoting a sequence.) There are $n!$ permutations of a set of n elements, since there are n ways to choose the first element of the sequence, $n - 1$ ways for the second element, $n - 2$ ways for the third, and so on.

A **k -permutation** of S is an ordered sequence of k elements of S , with no element appearing more than once in the sequence. (Thus, an ordinary permutation is an n -permutation of an n -set.) Here are the 2-permutations of the set $\{a, b, c, d\}$:

$ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db, dc$.

The number of k -permutations of an n -set is

$$n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!}, \quad (\text{C.1})$$

since there are n ways to choose the first element, $n - 1$ ways to choose the second element, and so on, until k elements are chosen, with the last element chosen from the remaining $n - k + 1$ elements. For the above example, with $n = 4$ and $k = 2$, the formula (C.1) evaluates to $4!/2! = 12$, matching the number of 2-permutations listed.

Combinations

A **k -combination** of an n -set S is simply a k -subset of S . For example, the 4-set $\{a, b, c, d\}$ has six 2-combinations:

ab, ac, ad, bc, bd, cd .

(Here we use the shorthand of omitting the braces around each subset.) To construct a k -combination of an n -set, choose k distinct (different) elements from the n -set. The order of selecting the elements does not matter.

We can express the number of k -combinations of an n -set in terms of the number of k -permutations of an n -set. Every k -combination has exactly $k!$ permutations of its elements, each of which is a distinct k -permutation of the n -set. Thus the number of k -combinations of an n -set is the number of k -permutations divided by $k!$. From equation (C.1), this quantity is

$$\frac{n!}{k!(n-k)!} . \quad (\text{C.2})$$

For $k = 0$, this formula tells us that the number of ways to choose 0 elements from an n -set is 1 (not 0), since $0! = 1$.

Binomial coefficients

The notation $\binom{n}{k}$ (read “ n choose k ”) denotes the number of k -combinations of an n -set. Equation (C.2) gives

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} .$$

This formula is symmetric in k and $n - k$:

$$\binom{n}{k} = \binom{n}{n-k} . \quad (\text{C.3})$$

These numbers are also known as *binomial coefficients*, due to their appearance in the *binomial theorem*:

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} , \quad (\text{C.4})$$

where $n \in \mathbb{N}$ and $x, y \in \mathbb{R}$. The right-hand side of equation (C.4) is called the *binomial expansion* of the left-hand side. A special case of the binomial theorem occurs when $x = y = 1$:

$$2^n = \sum_{k=0}^n \binom{n}{k}.$$

This formula corresponds to counting the 2^n binary n -strings by the number of 1s they contain: $\binom{n}{k}$ binary n -strings contain exactly k 1s, since there are $\binom{n}{k}$ ways to choose k out of the n positions in which to place the 1s.

Many identities involve binomial coefficients. The exercises at the end of this section give you the opportunity to prove a few.

Binomial bounds

You sometimes need to bound the size of a binomial coefficient. For $1 \leq k \leq n$, we have the lower bound

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1} \\ &= \left(\frac{n}{k}\right) \left(\frac{n-1}{k-1}\right) \cdots \left(\frac{n-k+1}{1}\right) \\ &\geq \left(\frac{n}{k}\right)^k. \end{aligned} \tag{C.5}$$

Taking advantage of the inequality $k! \geq (k/e)^k$ derived from Stirling's approximation (3.25) on page 67, we obtain the upper bounds

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1)\cdots(n-k+1)}{k(k-1)\cdots 1} \\ &\leq \frac{n^k}{k!} \\ &\leq \left(\frac{en}{k}\right)^k. \end{aligned} \tag{C.6}$$

For all integers k such that $0 \leq k \leq n$, you can use induction (see Exercise C.1-12) to prove the bound

$$\binom{n}{k} \leq \frac{n^n}{k^k (n-k)^{n-k}}, \tag{C.7}$$

where for convenience we assume that $0^0 = 1$. For $k = \lambda n$, where $0 \leq \lambda \leq 1$, we can rewrite this bound as

$$\begin{aligned}
\binom{n}{\lambda n} &\leq \frac{n^n}{(\lambda n)^{\lambda n} ((1-\lambda)n)^{(1-\lambda)n}} \\
&= \left(\left(\frac{1}{\lambda} \right)^\lambda \left(\frac{1}{1-\lambda} \right)^{1-\lambda} \right)^n \\
&= 2^{n H(\lambda)},
\end{aligned}$$

where

$$H(\lambda) = -\lambda \lg \lambda - (1-\lambda) \lg(1-\lambda) \quad (\text{C.8})$$

is the **(binary) entropy function** and where, for convenience, we assume that $0 \lg 0 = 0$, so that $H(0) = H(1) = 0$.

Exercises

C.1-1

How many k -substrings does an n -string have? (Consider identical k -substrings at different positions to be different.) How many substrings does an n -string have in total?

C.1-2

An n -input, m -output **boolean function** is a function from $\{0, 1\}^n$ to $\{0, 1\}^m$. How many n -input, 1-output boolean functions are there? How many n -input, m -output boolean functions are there?

C.1-3

In how many ways can n professors sit around a circular conference table? Consider two seatings to be the same if one can be rotated to form the other.

C.1-4

In how many ways is it possible to choose three distinct numbers from the set $\{1, 2, \dots, 99\}$ so that their sum is even?

C.1-5

Prove the identity

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1} \quad (\text{C.9})$$

for $0 < k \leq n$.

C.1-6

Prove the identity

$$\binom{n}{k} = \frac{n}{n-k} \binom{n-1}{k}$$

for $0 \leq k < n$.

C.1-7

To choose k objects from n , you can make one of the objects distinguished and consider whether the distinguished object is chosen. Use this approach to prove that

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

C.1-8

Using the result of Exercise C.1-7, make a table for $n = 0, 1, \dots, 6$ and $0 \leq k \leq n$ of the binomial coefficients $\binom{n}{k}$ with $\binom{0}{0}$ at the top, $\binom{1}{0}$ and $\binom{1}{1}$ on the next line, then $\binom{2}{0}$, $\binom{2}{1}$, and $\binom{2}{2}$, and so forth. Such a table of binomial coefficients is called *Pascal's triangle*.

C.1-9

Prove that

$$\sum_{i=1}^n i = \binom{n+1}{2}.$$

C.1-10

Show that for any integers $n \geq 0$ and $0 \leq k \leq n$, the expression $\binom{n}{k}$ achieves its maximum value when $k = \lfloor n/2 \rfloor$ or $k = \lceil n/2 \rceil$.

★ C.1-11

Argue that for any integers $n \geq 0, j \geq 0, k \geq 0$, and $j + k \leq n$,

$$\binom{n}{j+k} \leq \binom{n}{j} \binom{n-j}{k}. \quad (\text{C.10})$$

Provide both an algebraic proof and an argument based on a method for choosing $j + k$ items out of n . Give an example in which equality does not hold.

★ **C.1-12**

Use induction on all integers k such that $0 \leq k \leq n/2$ to prove inequality (C.7), and use equation (C.3) to extend it to all integers k such that $0 \leq k \leq n$.

★ **C.1-13**

Use Stirling's approximation to prove that

$$\binom{2n}{n} = \frac{2^{2n}}{\sqrt{\pi n}} (1 + O(1/n)). \quad (\text{C.11})$$

★ **C.1-14**

By differentiating the entropy function $H(\lambda)$, show that it achieves its maximum value at $\lambda = 1/2$. What is $H(1/2)$?

★ **C.1-15**

Show that for any integer $n \geq 0$,

$$\sum_{k=0}^n \binom{n}{k} k = n 2^{n-1}. \quad (\text{C.12})$$

★ **C.1-16**

Inequality (C.5) provides a lower bound on the binomial coefficient $\binom{n}{k}$. For small values of k , a stronger bound holds. Prove that

$$\binom{n}{k} \geq \frac{n^k}{4k!} \quad (\text{C.13})$$

for $k \leq \sqrt{n}$.

C.2 Probability

Probability is an essential tool for the design and analysis of probabilistic and randomized algorithms. This section reviews basic probability theory.

We define probability in terms of a *sample space* S , which is a set whose elements are called *outcomes* or *elementary events*. Think of each outcome as a possible result of an experiment. For the experiment of flipping two distinguishable coins, with each individual flip resulting in a head (H) or a tail (T), you can view the sample space S as consisting of the set of all possible 2-strings over $\{H, T\}$:

$$S = \{HH, HT, TH, TT\}.$$

An *event* is a subset¹ of the sample space S . For example, in the experiment of flipping two coins, the event of obtaining one head and one tail is $\{HT, TH\}$. The event S is called the *certain event*, and the event $\emptyset$ is called the *null event*. We say that two events A and B are *mutually exclusive* if $A \cap B = \emptyset$. An outcome s also defines the event $\{s\}$, which we sometimes write as just s . By definition, all outcomes are mutually exclusive.

Axioms of probability

A *probability distribution* $\Pr \{ \}$ on a sample space S is a mapping from events of S to real numbers satisfying the following *probability axioms*:

1. $\Pr \{A\} \geq 0$ for any event A .
2. $\Pr \{S\} = 1$.
3. $\Pr \{A \cup B\} = \Pr \{A\} + \Pr \{B\}$ for any two mutually exclusive events A and B . More generally, for any sequence of events $A_1, A_2, \dots$ (finite or countably infinite) that are pairwise mutually exclusive,

$$\Pr \left\{ \bigcup_i A_i \right\} = \sum_i \Pr \{A_i\} .$$